NM2b
Electrodynamics, python and MPI

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Abstract

Project for a term paper in "Numerical Methods in Classical Field Theory and Quantum Mechanics". In this project a wave equation is solved with the finite difference method.

1 Electrodynamics

The Maxwell equations are equivalent to a set of wave equations for the scalar potential ϕ and the vector potential $\vec{A}$. In the Lorentz gauge, the wave equations decouple, and e.g. the one for the scalar potential becomes

$$-\Delta\phi(\vec{x},t) + \frac{1}{c^2}\frac{\partial^2\phi(\vec{x},t)}{\partial t^2} = \frac{\rho(\vec{x},t)}{\epsilon_0}\,, \tag{1}$$

where c is the speed of light, ρ the charge density and ϵ_0 the vacuum permitivity. We simplify the problem to one spatial dimension and $\rho = 0$, and restrict ϕ to a finite volume $x \in [0, L]$. The problem to solve then is

$$\partial_t^2\phi = c^2\partial_x^2\phi\,, \tag{2}$$

with

$$\begin{aligned} \text{Boundary condition} \quad & \phi(0,t) = \phi(L,t) && = 0 && (3)\\ \text{Initial conditions} \quad & \phi(x,0) && = f(x) && (4)\\ \text{and} \quad & \partial_t\phi(x,0) && = g(x)\,. && (5) \end{aligned}$$

The initial conditions are given functions $f, g : [0, L] \to \mathbb{R}$.

2 Finite Differences

The derivatives can be approximated by

$$\partial_t^2 \phi = \frac{-2\phi(x,t) + \phi(x,t+\Delta_t) + \phi(x,t-\Delta_t)}{\Delta_t^2} + O(\Delta_t^2) \quad (6)$$

$$\partial_x^2 \phi = \frac{-2\phi(x,t) + \phi(x+\Delta_x,t) + \phi(x-\Delta_x,t)}{\Delta_x^2} + O(\Delta_x^2) \quad (7)$$

$$\partial_t \phi = \frac{\phi(x,t+\Delta_t) - \phi(x,t)}{\Delta_t} + O(\Delta_t) \quad (8)$$

$$(9)$$

Therefore the discretized wave equation can be brought into the form

$$\phi(x,t+\Delta_t) = \frac{c^2\Delta_t^2}{\Delta_x^2}\left[-2\phi(x,t) + \phi(x+\Delta_x,t) + \phi(x-\Delta_x,t)\right] + 2\phi(x,t) - \phi(x,t-\Delta_t)\,. \quad (10)$$

If ϕ is known for all x at times t and $t-\Delta_t$, this equation allows it to be evolved to time $t+\Delta_t$. This method is known to be stable if $\Delta_t \leq \frac{\Delta_x}{c}$. To start the iteration, the initial two time-slices of the solution are required. The slice at $t=0$ is given by one of the initial conditions, $\phi(x,t) = f(x)$. The next slice can be obtained from the other initial condition, together with the discretized time derivative: $\phi(x,\Delta_t) = f(x) + \Delta_t g(x)$.

3 Parallelization

The problem can be solved by N_{proc} processes in parallel. To this aim the interval $[0,L]$ is split into N_{proc} equal pieces, and every process contains one of the pieces and computes its part of eq.(10). To do so for its inner boundary points, it needs to obtain outer boundaries from the neighboring processes.

4 Task

Write a function that solves the 1+1 dimensional wave equation as outlined above. The program should allow for arbitrary functions $f(x)$ and $g(x)$.

- Write a sequential version (non-parallelized)
- Try to find a solvable example in the continuum, and work out the solution. Demonstrate, that your code obtains the correct solution and that the discretization errors behave as expected (vary Δ_x, Δ_t).
- Parallelize your code with MPI. Show that sequential and parallel versions give identical results.

- Add plotting capabilities (either directly, or by writing ϕ configurations to files and plotting with an external program).
- In your term paper of about 10 pages
  - Give a short theoretical introduction to the problem
  - Demonstrate that your implementation is correct.
  - Demonstrate that your parallelization is efficient. Show a strong scaling plot.

References

[1] `https://moodle.uni-wuppertal.de/course/view.php?id=29658`